Baseline Notebook

Setup Environment

In [1]:

```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
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0:00
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the follo
wing dependency conflicts.
umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.
5.2 which is incompatible.
hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2
which is incompatible.
Mounted at /content/gdrive

You can now save your data files in: /content/gdrive/MyDrive/36106/assignm
ent/AT3/data
```

Student Information

In [2]:
```python
group_name = "20"
student_name = "Shanu Sharma"
student_id = "25979360"
```

In [3]:
```python
# Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

group_name

20

In [4]:
```python
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_name', value=student_name)
```

student_name

Shanu Sharma

In [5]:
```python
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_id', value=student_id)
```

student_id

25979360

---

0. Python Packages

0.a Install Additional Packages

> If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

In [6]:
```python
# No additional packages are required.
```

0.b Import Packages

In [7]:
```python
import pandas as pd
```

A. Assess Baseline Model

In [8]:
```python
# DO NOT MODIFY THE CODE IN THIS CELL
# Load data
```

```python
try:
  X_train = pd.read_csv(at.folder_path / 'X_train.csv')
  y_train = pd.read_csv(at.folder_path / 'y_train.csv')

  X_val = pd.read_csv(at.folder_path / 'X_val.csv')
  y_val = pd.read_csv(at.folder_path / 'y_val.csv')

  X_test = pd.read_csv(at.folder_path / 'X_test.csv')
  y_test = pd.read_csv(at.folder_path / 'y_test.csv')
except Exception as e:
  print(e)
```

```
No columns to parse from file
```

A.1 Baseline Applicability

In [9]:
```python
baseline_applicability_explanations = """
This use case implements unsupervised product clustering. There is no tar

A supervised baseline requires labelled training examples: a model is fit

Cluster quality cannot be assessed using supervised metrics such as accur
"""
```

In [10]:
```python
print_tile(size="h3", key="baseline_applicability_explanations", value=ba
```

baseline_applicability_explanations

This use case implements unsupervised product clustering. There is no target variable in the source data — no ground-truth segment label, class, or outcome column exists that products could be predicted against. A supervised baseline requires labelled training examples: a model is fitted to X_train and evaluated on how accurately it predicts y_val or y_test. Neither condition holds here. y_train, y_val, and y_test are empty placeholder files saved by the preparation notebook to satisfy the standard workflow structure; they contain no labels because none exist for this task. Cluster quality cannot be assessed using supervised metrics such as accuracy, F1, or RMSE. Instead it is measured using internal validity metrics — silhouette score, Calinski-Harabasz index, and Davies-Bouldin index — which evaluate the compactness and separation of discovered clusters without requiring ground-truth labels. These assessments are conducted in the clustering notebook, where candidate solutions across multiple algorithms and parameter settings can be compared before the final model is selected.

A.2 Performance Metric Applicability

In [11]:
```python
performance_metrics_explanations = """
A target-based baseline metric is not selected in this notebook because t
```

```
Reporting an arbitrary clustering baseline would be misleading because cl
"""
```

In [12]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='performance_metrics_explanations', value=perfo
```

performance_metrics_explanations

A target-based baseline metric is not selected in this notebook because this is an unsupervised clustering task. There is no ground-truth product segment, class label, or target variable to compare predictions against, and the y_train, y_val, and y_test files are only template placeholders. Reporting an arbitrary clustering baseline would be misleading because cluster quality depends on the selected feature space, scaling, distance assumptions, model family, and comparison across candidate cluster solutions. The appropriate evaluation belongs in the final clustering notebook, where candidate models can be compared using clustering-specific evidence such as silhouette score, Davies-Bouldin score, Calinski-Harabasz score, inertia or elbow behaviour, cluster balance, and business interpretability.

A.3 Baseline Model Performance

In [13]:
```
baseline_performance_explanations = f"""
No baseline model performance is reported because no baseline clustering

Fitting a single arbitrary clustering solution here would start the actua
"""
```

In [14]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='baseline_performance_explanations', value=base
```

baseline_performance_explanations

No baseline model performance is reported because no baseline clustering model is fitted in this notebook. This is intentional for the selected unsupervised product clustering use case. The prepared data contains 254 sales-active products and 20 product-level features describing sales, pricing, cost, discount, promotion, and estimated margin behaviour. Fitting a single arbitrary clustering solution here would start the actual modelling experiment without comparing alternatives. That decision belongs in the final clustering model training notebook, where candidate cluster solutions can be assessed together before selecting and interpreting the final product groups. This keeps the baseline notebook focused on explaining why target-based baseline evaluation is not applicable, while preserving model selection for the correct stage of the workflow.